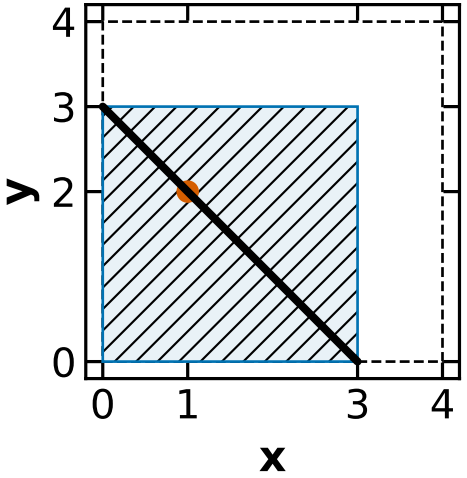

Dashed box: original bounds $[0,4]^2$

Solid line: original row $x + y = 3$

Hatched box: bounds $[0,3]^2$

Marked feasible point: (1,2)